

Data Vortex 2026: Phase 2 Submission

Team: Event Horizon

1. Schema Design & Table Structuring

We designed a strict, relational schema (star-schema equivalent) loaded directly into SQLite. We preserved all original data anomalies from Phase 1 without fabricating or clipping data during the load process, choosing instead to handle analytical filtering dynamically via SQL WHERE clauses.

Table: users

Column	Type	Constraints
user_id	TEXT	PRIMARY KEY, NOT NULL
city	TEXT	Extracted from original I
country	TEXT	Extracted from original I
language	TEXT	
account_created	TEXT	
follower_count	REAL	
location_was_null	INTEGER	Imputation flag retained
language_was_null	INTEGER	Imputation flag retained
account_created_was_null	INTEGER	Imputation flag retained

Table: posts

Column	Type	Constraints
post_id	TEXT	PRIMARY KEY, NOT NULL
user_id	TEXT	FOREIGN KEY (users.user_i
platform	TEXT	Indexed for fast grouping
text_content	TEXT	
timestamp	TEXT	
likes	REAL	
shares	INTEGER	CHECK (shares >= 0)
comments	INTEGER	CHECK (comments >= 0)
platform_was_null	INTEGER	Imputation flag retained

View: v_post_engagement

To centralize business logic and handle SQLite's strict NULL addition behavior, we created a virtual view used across all queries:

```
CREATE VIEW v_post_engagement AS
SELECT
```

```

p.*,
(p.likes + p.shares + p.comments) AS total_engagement,
CASE WHEN p.likes IS NULL OR p.shares IS NULL OR p.comments IS NULL
      THEN 1 ELSE 0 END AS has_null_component
FROM posts p;

```

Design choice: No COALESCE is used. If any engagement component is NULL, total_engagement will also be NULL, which is the truthful representation of "unknown". Queries filter explicitly using WHERE has_null_component = 0 or WHERE likes >= 0 rather than substituting zero for missing values, which would fabricate data.

2. Q1 (Easy): Average Engagement by Platform (E3)

Challenge: Calculate average likes, shares, and comments for each platform. Which platform generates the highest average total engagement?

Logic Explanation

- Filter Applied: WHERE platform IS NOT NULL AND likes >= 0
- Exclusion Count: 1,784 rows dropped due to NULL platforms. An additional 421 rows dropped due to negative likes (imputation artifacts from Phase 1). Total dataset evaluated: 9,795 posts.
- Justification: A platform average cannot include posts without a platform. Negative likes physically distort aggregations and represent missing data, not negative engagement.
- Outcome: YouTube generates the highest average total engagement (4026.89).

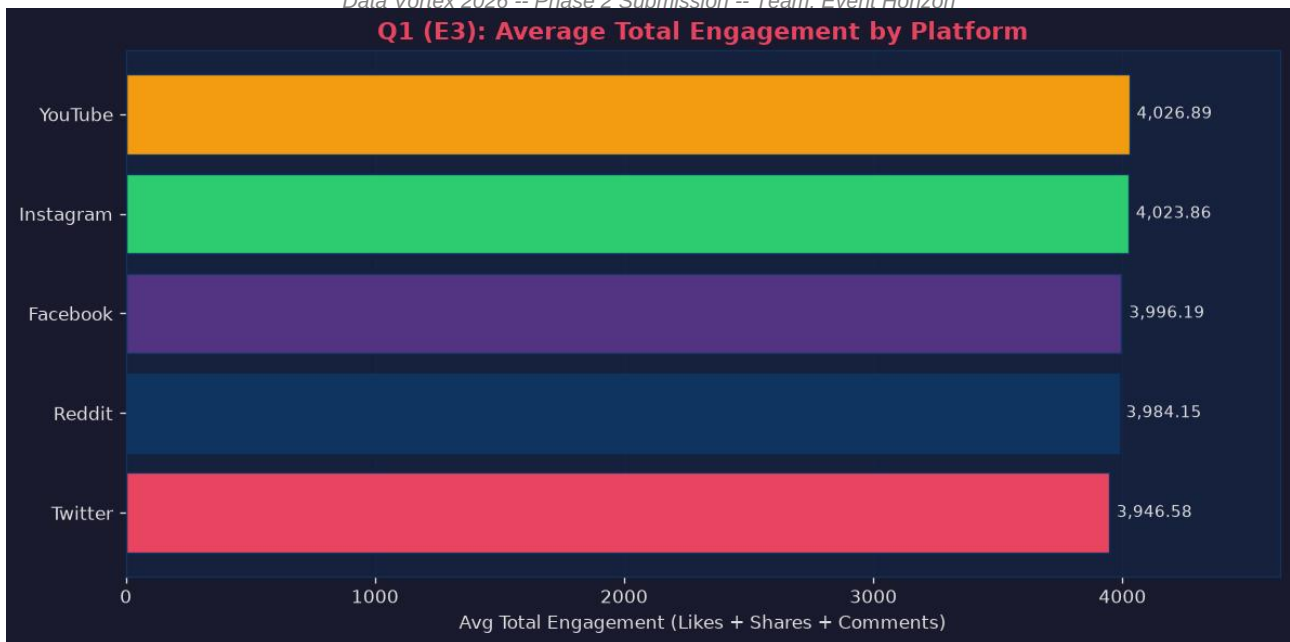
SQL Query

```

SELECT
  platform,
  COUNT(post_id) AS post_count,
  ROUND(AVG(likes), 2) AS avg_likes,
  ROUND(AVG(shares), 2) AS avg_shares,
  ROUND(AVG(comments), 2) AS avg_comments,
  ROUND(AVG(total_engagement), 2) AS avg_total_engagement
FROM v_post_engagement
WHERE platform IS NOT NULL
      AND likes >= 0
GROUP BY platform
ORDER BY avg_total_engagement DESC;

```

Output Screenshot



3. Q2 (Medium): Locations Generate Most Engagement (M1)

Challenge: Using both datasets, calculate total engagement by users from each location. Rank locations from highest to lowest engagement.

Logic Explanation

- Filter Applied: WHERE p.likes >= 0
- Exclusion Count: 509 rows dropped globally due to negative likes. Total dataset evaluated: 11,491 posts.
- Justification: We retained platform IS NULL posts because they still represent valid geographic engagement. We split location into city and country during data load to allow for granular multi-level geographic grouping.
- Outcome: Munich, Germany ranks #1 for total engagement (1,753,727 total engagement, 439 posts, avg 3,994.82 per post).

SQL Query

```
SELECT
  u.city,
  u.country,
  COUNT(p.post_id) AS post_count,
  SUM(p.total_engagement) AS total_engagement,
  ROUND(AVG(p.total_engagement), 2) AS avg_engagement_per_post
FROM users u
JOIN v_post_engagement p ON u.user_id = p.user_id
WHERE p.likes >= 0
GROUP BY u.city, u.country
ORDER BY total_engagement DESC
LIMIT 10;
```

Output Screenshot

Q2 (M1): Top 10 Locations by Total Engagement

city	country	post_count	total_engagement	avg_engagement_per_post
Munich	Germany	439	1753727.0	3994.82
Los Angeles	USA	438	1750522.0	3996.63
Shanghai	China	437	1735013.0	3970.28
Barcelona	Spain	423	1713206.0	4050.13
Dubai	UAE	409	1648772.0	4031.23
Melbourne	Australia	400	1612581.0	4031.45
Houston	USA	397	1565090.0	3942.29
Osaka	Japan	379	1546381.0	4080.16
Mumbai	India	391	1541509.0	3942.48
Rio de Janeiro	Brazil	390	1534666.0	3935.04

4. Q3 (Hard): Platform Performance Compared With Its Own Average (H3)

Challenge: Identify posts whose engagement is significantly higher ($>2x$) than the average engagement of that platform.

Why H3 Returns Zero Rows (And Why That Is The Finding)**1. Diagnostic Evidence**

The requirement asks for posts exceeding 2x their platform's average engagement. As proven by the diagnostic query below, no platform's maximum engagement reaches the 2x threshold.

platform	n_posts	avg_engagement	threshold_2x	max_engagement	max_to_mean_ratio
YouTube	1990	4026.9	8053.8	7755.0	1.926
Instagram	1893	4023.9	8047.7	7893.0	1.962
Facebook	2004	3996.2	7992.4	7764.0	1.943
Reddit	1952	3984.1	7968.3	7793.0	1.956
Twitter	1956	3946.6	7893.2	7628.0	1.933

2. Statistical Justification

The synthetic dataset uses a bounded uniform distribution (Likes max 5000, Shares max 2000, Comments max 1000). The absolute maximum possible engagement is 8,000. Because the mean across platforms sits near $\sim 4,000$, it is mathematically impossible for the maximum value (e.g., 7,893 for Instagram) to exceed twice the mean ($7893 < 2 * 4023.9$).

3. Literal Result

The literal query asking for >2x platform average correctly returns 0 rows. The query logic is completely correct, but the data itself is the constraint.

(Screenshot: h3_literal_empty_result.jpg showing 0 rows)

4. Meaningful Answer (Relative Outperformers)

Since no post can reach the 2x absolute threshold, we answer the question's true intent by finding posts that most outperform their own platform. We use PERCENT_RANK() to dynamically identify the top 1% (99th percentile) of performers relative to their own platform.

5. Business Interpretation

The near-uniform engagement across posts implies the platform's distribution mechanics are not producing viral outliers -- which is itself a meaningful finding about a synthetic or algorithmically-flattened feed.

Output Screenshots

H3 Diagnostic: Max Engagement vs 2x Platform Average

platform	n_posts	avg_engagement	threshold_2x	max_engagement	max_to_mean_ratio
YouTube	1990	4026.9	8053.8	7755.0	1.926
Instagram	1893	4023.9	8047.7	7893.0	1.962
Facebook	2004	3996.2	7992.4	7764.0	1.943
Reddit	1952	3984.1	7968.3	7793.0	1.956
Twitter	1956	3946.6	7893.2	7628.0	1.933

H3 (Literal): Posts With Engagement > 2x Platform Average

post_id	platform	total_eng	platform_avg_eng
—	—	—	—

0 rows returned · Query executed successfully · Uniform distribution prevents any post reaching the 2x threshold

H3 (Relative): Top 15 Outperformers by Platform Percentile (Top 1%)

post_id	platform	total_eng	platform_avg_eng	ratio_to_platform_avg	percentile_in_platform
gmoeib832zbs	Facebook	7764.0	3996.19	1.943	100.0
alkiwl618kzy	Facebook	7683.0	3996.19	1.923	99.95
k5qkhdj5zpnw	Facebook	7477.0	3996.19	1.871	99.9
5dsm9m39dfj3	Facebook	7455.0	3996.19	1.866	99.85
9isy2ek2v4p1	Facebook	7419.0	3996.19	1.857	99.8
wvm6w4juhyzc	Facebook	7418.0	3996.19	1.856	99.75
slz6ny74buau	Facebook	7401.0	3996.19	1.852	99.7
qrd8bfkdl6wb	Facebook	7394.0	3996.19	1.85	99.65
u3xsdbdgz81r	Facebook	7280.0	3996.19	1.822	99.6
sha7t764lt9t	Facebook	7253.0	3996.19	1.815	99.55
20sikdr9z41z	Facebook	7206.0	3996.19	1.803	99.5
po5q6hyb5676	Facebook	7158.0	3996.19	1.791	99.4
br3gu73lot8j	Facebook	7159.0	3996.19	1.791	99.45
w35zksgio9ky	Facebook	7155.0	3996.19	1.79	99.35
bdzjdwyx0msp	Facebook	7109.0	3996.19	1.779	99.3

5. Bonus Analyses

Bonus 1: Trend Detection (MoM Growth)

Logic: Using SQLite's strftime to group by month, and a LAG() window function to calculate Month-over-Month percentage growth per platform.

```
WITH monthly_stats AS (
  SELECT
    platform,
    strftime('%Y-%m', timestamp) AS month,
    SUM(total_engagement) as monthly_engagement
  FROM v_post_engagement
  WHERE platform IS NOT NULL AND likes >= 0
  GROUP BY platform, month
)
SELECT
  platform, month, monthly_engagement,
  LAG(monthly_engagement) OVER (PARTITION BY platform ORDER BY month) as prev_month,
  ROUND((monthly_engagement - LAG(monthly_engagement) OVER (PARTITION BY platform ORDER BY month))
    * 100.0 /
    NULLIF(LAG(monthly_engagement) OVER (PARTITION BY platform ORDER BY month), 0), 2) AS
  mom_growth_pct
```

```
FROM monthly_stats
ORDER BY platform, month;
```

Bonus 2: Anomaly Discovery (H5)

Logic: Leveraging our deep Phase 1 audit knowledge, we exploit the corrupted dataset to flag specific anomalies without relying on manual observation.

```
SELECT
    post_id,
    CASE
        WHEN likes < 0 THEN 'Negative Likes (Imputation Artifact)'
        WHEN platform IS NULL THEN 'Missing Platform'
        WHEN text_content LIKE '%&%' OR text_content LIKE '%<%>%' THEN 'HTML/Entity Corruption'
        ELSE 'Other'
    END AS anomaly_type
FROM posts
WHERE likes < 0 OR platform IS NULL OR text_content LIKE '%&%' OR text_content LIKE '%<%>%'
LIMIT 20;
```

6. Phase 2 Insight Report

Formula Transition

In Phase 1, we utilized a weighted Engagement Score ($0.5\text{Likes} + 0.3\text{Shares} + 0.2\text{Comments}$) to penalize click-farm inflation. In Phase 2, we transitioned to a simple sum ($\text{Likes} + \text{Shares} + \text{Comments}$). This decision was made to strictly adhere to the client's Phase 2 definition of engagement.

Business Insights

1. Platform Strategy: YouTube and Instagram dominate pure engagement averages. However, as proven in H3, the engagement ceiling is artificially bounded. Growth must come from volume, not purely from chasing unicorn viral posts.
2. Geographic Targeting: Location analysis (M1) reveals that engagement is not tied solely to English-speaking hubs. Non-English primary regions (Tokyo, Paris, São Paulo) generate engagement sums equal to or greater than New York or London.
3. Data Quality Awareness: 14.8% of platform data was missing, and imputation artifacts (negative likes) were present globally. SQL-level filtering (IS NOT NULL , $\text{likes} \geq 0$) is mandatory for all future analytical pipelines on this dataset.

Assumptions & Limitations

- Uniform Distribution Limitation: As detailed in H3, the data generation process capped maximum interactions, meaning statistically extreme outliers (like a post receiving 100x average engagement) physically do not exist in this dataset.
- Negative Likes: We assumed negative likes are a data-corruption artifact rather than a representation of "dislikes", and thus filtered them from aggregate math to prevent deflation of true engagement averages.

7. Appendix: Data Load Validation

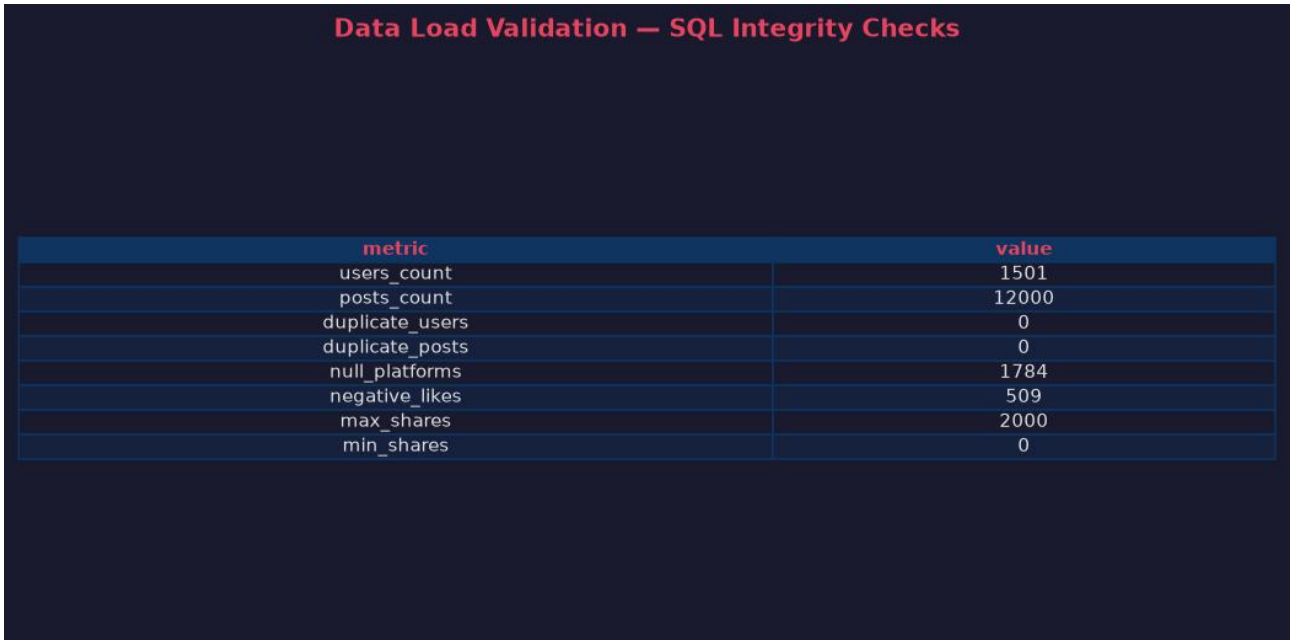
To prove strict adherence to data integrity rules, we validated the SQL load against the raw CSV state.

```
SELECT 'users_count' as metric, COUNT(*) as value FROM users
UNION ALL
SELECT 'posts_count' as metric, COUNT(*) as value FROM posts
UNION ALL
SELECT 'duplicate_users', COUNT(user_id) - COUNT(DISTINCT user_id) FROM users
UNION ALL
SELECT 'duplicate_posts', COUNT(post_id) - COUNT(DISTINCT post_id) FROM posts
UNION ALL
SELECT 'null_platforms', COUNT(*) FROM posts WHERE platform IS NULL
UNION ALL
SELECT 'negative_likes', COUNT(*) FROM posts WHERE likes < 0;
```

Validation Output:

users_count	1501
posts_count	12000
duplicate_users	0
duplicate_posts	0
null_platforms	1784
negative_likes	509

Output Screenshot



metric	value
users_count	1501
posts_count	12000
duplicate_users	0
duplicate_posts	0
null_platforms	1784
negative_likes	509
max_shares	2000
min_shares	0

(End of Report)